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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	AllStar
Selected Domain	Domain 2 - ALPHA RED
Selected SDG Track (1–17)	8 Decent Work and Economic Growth
Project / Solution Name	Orbyte Smart SME
Name (Leader)	PHAN YI YAN
Email & Phone (Leader)	phanyiyan@gmail.com & 011-12104025
University Affiliation(s)	Universiti Sains Malaysia (USM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small and Medium Enterprises (SMEs) constitute over 97% of business establishments in Malaysia, serving as vital economic drivers across manufacturing, retail and food industries. However, SMEs face significant operational inefficiencies and resource waste due to a lack of accessible, real-time tracking tools. Traditional inventory management relies on manual record-keeping or fragmented systems, leading to over-purchasing, material degradation, improper waste disposal and undetected utility leaks.

This operational blind spot directly contributes to rising business overhead and heightened environmental degradation, including increased landfill waste and carbon footprint. Furthermore, tight profit margins prevent SMEs from adopting expensive enterprise resource planning (ERP) systems. The lack of actionable operational insights forces business owners to make reactive rather than predictive decisions, resulting in avoidable material losses and sub-optimal resource allocation. Addressing this challenge is crucial to reduce SME operational waste directly enhances economic resilience while advancing environmental sustainability goals across local supply chains.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

The core solution leverages generative AI (Google Gemini API) to transform raw inventory, waste logs and usage trends into actionable optimization strategies. Instead of complex data tables, the AI engine intentionally processes material inputs to automatically generate predictive waste reduction alerts, optimal reordering thresholds and sustainable operational recommendations customized to the business context.

To build a fully functional prototype, our team leverages key technical skills from the SDC Phase 1 workshops:

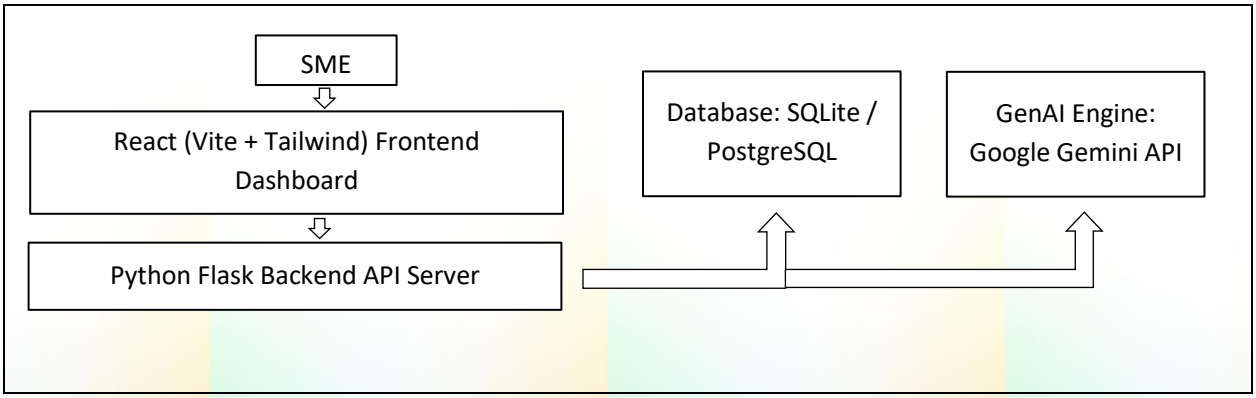
1. GenAI APIs: Integrating Google Gemini API for natural language reasoning, waste classification, and operational recommendations.
2. Database Architecture & Data Pipelines: Structuring relational database models to track daily material throughput, waste categories, and cost impact.
3. Cloud Deployment: Containerizing and deploying the Flask backend and React frontend (via Render/Ngrok) to deliver a live, accessible web dashboard for real-time demonstration.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We applied design-thinking methodology by conducting user needs and process reviews with SME operational workflows. We identified that SME operators require zero-friction data entry and rapid, actionable insights rather than disorganized data tables.

To ensure technical viability and Phase 2 deployment readiness, the system prioritizes a modular architecture: the React frontend provides lightweight visual analytics while the Flask backend abstracts complex prompt engineering and data validation. Fallback mechanisms and cached API responses ensure dashboard stability during network latency. By Phase 2, the prototype will be fully deployed with pre-populated sample datasets and live interactive inputs for judges to test real-time AI reasoning.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.2 "Productivity through technological upgrading and innovation" Target 8.3 "Support for entrepreneurship and the growth of micro, small and medium-sized enterprises"
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	According to global and national SME operational studies, unmonitored small-to-medium enterprise operations experience an estimated 25% to 30% material and inventory waste due to over-purchasing, unoptimized scheduling, and manual tracking errors. *(Source: World Bank SME Sustainability Benchmarks & UNEP Food Waste Index Report).*
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage reduction in total operational waste (kg/month) and percentage improvement in inventory utilization efficiency.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	A 20% overall reduction in preventable operational waste and a 15% reduction in material tracking overhead simulated across SME test workflows.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Gemini	Gemini 1.5 Flash/Pro	Google AI Terms of Service

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



PHAN YI YAN (Group Leader)

25/10/2026